

HAMILTON, TX, USA

WMO: 722231

Lat: 31.666N

Lon: 98.149W

Elev: 1300

StdP: 14.02

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00357

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.1	27.6	9.3	9.3	34.2	13.7	11.6	37.9	29.4	57.2	26.4	54.2	10.0	10	0.502

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.4	101.7	70.7	99.3	71.3	97.2	71.8	76.8	85.6	76.0	85.3	75.2	85.0	9.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.4	139.5	77.0	74.4	135.0	76.7	73.2	129.5	76.6	41.2	85.5	40.3	84.2	39.7	84.8	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
24.9	22.0	19.6	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			15.9	104.2	2.4	3.6	14.2	106.7	12.8	108.9	11.4	110.9	9.7	113.5
			14.3	78.1	2.3	1.3	12.7	79.0	11.3	79.8	10.0	80.5	8.4	81.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	66.9	47.9	51.9	59.0	66.1	73.5	81.6	84.5	84.8	77.8	67.9	57.6	49.3
	DBStd	15.28	9.61	11.31	9.48	7.61	6.39	3.96	3.40	4.64	5.91	8.52	9.39	9.85
	HDD50	465	153	103	32	1	0	0	0	0	0	4	37	135
	HDD65	2049	533	380	230	78	13	0	0	0	1	70	251	493
	CDD50	6631	88	157	310	483	728	949	1069	1078	833	559	264	113
	CDD65	2740	3	14	43	110	276	499	604	613	385	159	28	6
	CDH74	31445	56	122	398	1011	2520	5774	7760	7982	4041	1502	218	61
	CDH80	16099	7	22	87	312	985	3010	4436	4656	1996	539	42	7
Wind	WSAvg	9.6	9.9	10.9	11.3	11.7	10.5	9.7	8.4	7.7	7.4	8.9	9.3	9.6
Precipitation	PrecAvg	33.5	1.9	2.6	2.8	3.0	3.7	4.0	2.2	2.7	2.9	3.5	2.3	1.9
	PrecMax	49.4	4.4	8.0	7.8	7.4	9.5	11.0	6.1	8.9	5.9	7.0	8.2	9.1
	PrecMin	19.8	0.4	0.0	0.1	0.5	0.8	1.0	0.0	0.0	0.5	0.6	0.0	0.2
	PrecStd	8.0	1.0	1.9	1.7	1.8	2.2	2.6	1.7	2.4	1.6	2.0	2.1	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.3	82.4	86.3	92.7	94.4	100.4	104.8	104.5	100.3	93.9	84.9	80.8
		MCWB	58.2	59.2	62.1	66.6	71.8	71.9	70.7	70.3	68.3	68.4	66.8	61.2
	2%	DB	74.5	77.4	82.2	87.0	91.2	98.0	100.5	102.1	96.8	88.9	79.6	74.7
		MCWB	56.3	61.0	62.4	66.3	70.7	72.0	70.8	71.0	70.3	68.7	63.4	59.5
	5%	DB	68.9	73.3	78.9	83.2	88.6	95.4	98.6	99.8	93.3	85.9	75.6	69.9
		MCWB	55.3	58.8	61.7	65.8	70.4	72.4	71.4	71.6	70.1	67.6	62.8	58.1
	10%	DB	63.8	69.7	74.6	80.0	86.0	93.1	96.3	97.9	90.8	82.4	72.3	65.6
		MCWB	51.9	58.2	60.0	64.9	69.8	72.4	72.1	71.9	69.7	65.1	61.0	57.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.6	68.4	69.0	73.2	76.7	78.0	77.8	77.2	77.0	75.1	72.0	67.9
		MCDB	70.2	72.9	75.6	80.7	84.1	88.2	86.3	86.0	83.2	80.4	77.0	71.1
	2%	WB	61.2	66.3	67.0	71.1	75.0	76.4	76.6	76.1	75.5	73.6	69.1	65.0
		MCDB	68.4	71.7	74.3	79.1	83.7	86.0	85.9	86.5	82.9	79.2	72.4	69.8
	5%	WB	57.5	64.0	65.6	69.5	73.7	75.5	75.7	75.2	74.6	72.3	66.4	62.0
		MCDB	64.4	69.1	73.1	78.8	82.2	84.9	85.7	86.7	82.5	78.9	70.4	65.7
	10%	WB	54.3	60.8	63.5	68.2	72.7	74.7	74.9	74.4	73.4	70.6	64.3	58.3
		MCDB	62.6	66.9	71.0	75.8	81.0	84.6	85.6	86.4	82.1	77.9	69.6	64.9
Mean Daily Temperature Range	5% DB	MDBR	22.5	22.0	23.2	22.7	20.5	21.4	22.8	22.4	21.8	22.7	22.1	21.1
		MCDBR	29.6	26.8	26.4	26.4	23.5	24.4	25.7	25.7	25.0	25.0	24.6	27.1
		MCWBR	15.7	15.1	11.8	12.3	8.2	5.2	5.7	5.6	6.9	10.0	12.3	15.7
	5% WB	MCDBR	24.6	21.6	21.0	19.8	19.0	19.8	21.1	20.8	19.2	19.4	18.8	20.9
		MCWBR	16.9	15.2	12.2	11.3	8.2	5.8	5.6	5.4	5.9	10.9	13.0	16.6
Clear-Sky Solar Irradiance	taub	0.305	0.315	0.341	0.380	0.413	0.414	0.425	0.416	0.389	0.356	0.328	0.310	
	taud	2.520	2.500	2.440	2.355	2.320	2.355	2.366	2.372	2.393	2.471	2.488	2.512	
	Ebn at Noon	294	301	298	289	279	277	273	275	279	282	282	285	
	Edh at Noon	27	30	35	39	41	40	39	38	36	31	28	26	
All-Sky Solar Radiation	RadAvg	970	1154	1488	1795	1891	2143	2164	1990	1667	1328	1019	858	
	RadStd	103	176	125	162	183	175	160	129	140	200	128	116	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (48 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page